

DS 116 Data Visualization

Single numeric variable

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Section 1

Histogram

Movies data set

```

movies <- read.csv('Data/movies.csv', stringsAsFactors = F)
str(movies)

## 'data.frame':    2911 obs. of  32 variables:
## $ title          : chr  "Zoom" "Zoolander 2" "Zombieland" "Zodiac"
## $ genre_first    : chr  "Action" "Comedy" "Adventure" "Crime" ...
## $ year           : int   2006 2016 2009 2007 1998 2012 2005 2008 199
## $ duration       : int   83 102 88 162 116 157 101 101 119 90 ...
## $ gross_adjusted : num   14142117 29451448 86365946 39077724 2978040
## $ budget_adjusted : int   42555556 51065177 26964263 76858659 7519018
## $ gross          : int   11631245 28837115 75590286 33048353 1980338
## $ budget         : int   35000000 50000000 23600000 65000000 5000000
## $ cast_facebook_likes: int   5022 24107 28011 36928 1209 2759 32232 638
## $ reviews        : int   176 376 998 966 232 1198 338 490 709 297 ..
## $ index           : num   1.22 1.02 1.14 1.18 1.5 ...
## $ Rated           : chr   "PG" "PG-13" "R" "R" ...
## $ Genre           : chr   "Action, Adventure, Comedy" "Comedy" "Adven
## $ Writer          : chr   "Adam Rifkin (screenplay), David Berenbaum
## $ Actors          : chr   "Tim Allen, Courteney Cox, Chevy Chase, Spe
## $ Plot            : chr   "Former superhero Jack is called back to wo
## $ Language        : chr   "English" "English, Italian, Spanish" "Engl

```

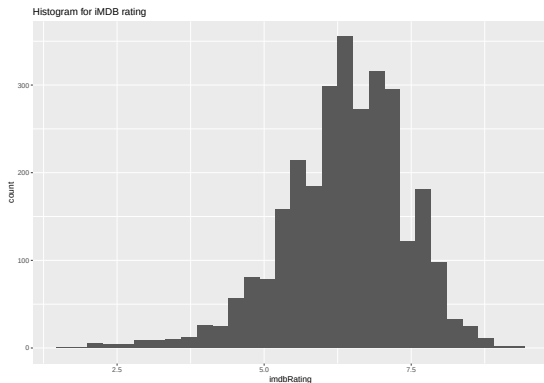
Histogram

- A histogram displays the frequency and distribution for a range of quantitative groups.
- Bar charts compare quantities for different categories, a histogram technically compares the number of observations across a range of value 'bins' using the size of lines/bars to represent the quantitative counts.
- Histogram allows to understand the shape of the distribution of the data

Doing with ggplot

- You need to specify only one aesthetics: x
- Use `geom_histogram` as a geometric object

```
ggplot(data = movies, aes(x = imdbRating)) +  
  geom_histogram() + ggtitle('Histogram for iMDB rating')
```



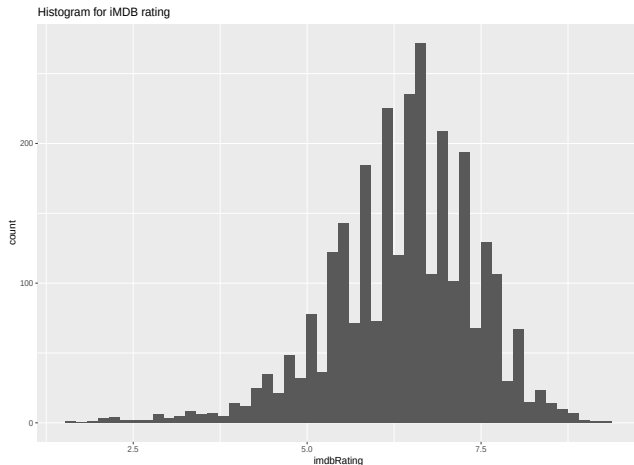
Doing with ggplot2

- in ggplot you can either specify the number of bins (bins) or bin width (binwidth). One can be derived from another.
- The default value for number of bins is 30.
- There is no one golden rule on choosing number of bins, however in general
 - More bins (smaller binwidth) will result in higher detalization
 - Less bins (larger binwidth) will result in lower detalization

Doing with ggplot

Number of bins = 50

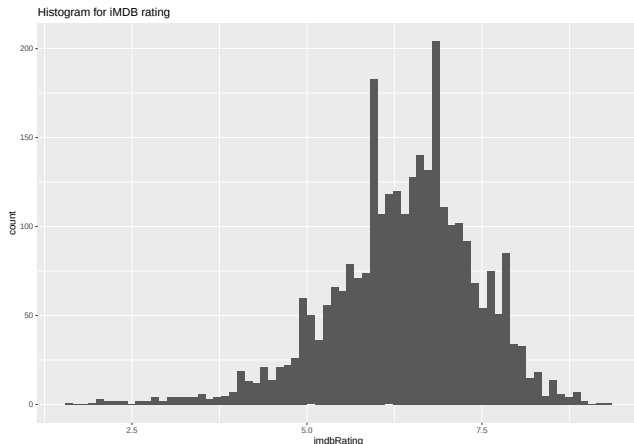
```
ggplot(data = movies, aes(x = imdbRating)) + geom_histogram(bins = 50) +  
  ggtitle('Histogram for iMDB rating')
```



Doing with ggplot

Number of bins = 70

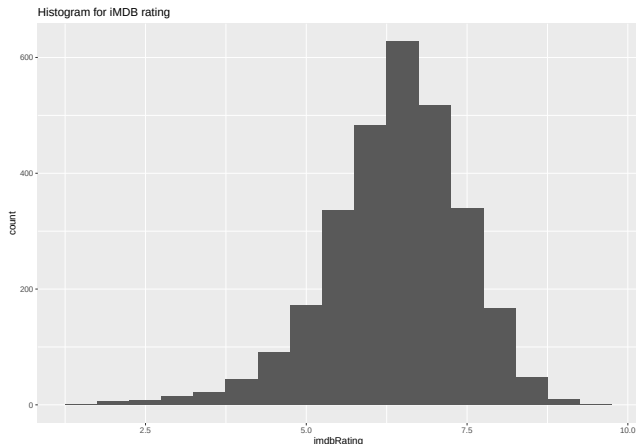
```
ggplot(data = movies, aes(x = imdbRating)) +  
  geom_histogram(bins = 70) + ggtitle('Histogram for iMDB rating')
```



Doing with ggplot

Alternatively you can set up the binwidth: $\text{binwidth} = 0.5$

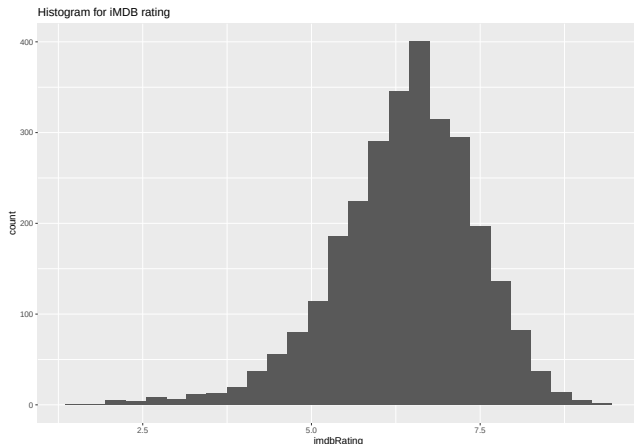
```
ggplot(data = movies, aes(x = imdbRating)) +  
  geom_histogram(binwidth = 0.5) + ggtitle('Histogram for iMDB rating')
```



Doing with ggplot

Binwidth = 0.3

```
ggplot(data = movies, aes(x = imdbRating)) + geom_histogram(binwidth = 0.3)  
  ggtitle('Histogram for iMDB rating')
```



Choosing the number of bins

- There is no one golden rule on how many bins need to be there
- Do try and error until you get histogram that can be interpreted
- However, there are few approaches for the calculation of optimal number of bins

Methods for choosing the number of bins

- 1 Square root choice

$$k = \lceil \sqrt{n} \rceil$$

- 2 Rice rule

$$k = \lceil 2\sqrt[3]{n} \rceil$$

- 3 Sturges' formula

$$k = \lceil \log_2 n \rceil + 1$$

Methods for choosing the binwidth

- 1 Scott's normal reference rule (when data is approximately normal)

$$h = \frac{3.49\hat{\sigma}}{\sqrt[3]{n}}$$

- 2 Freedman–Diaconis' rule - a variation of Scott's rule but less sensitive to outliers

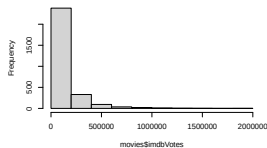
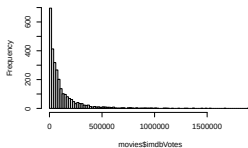
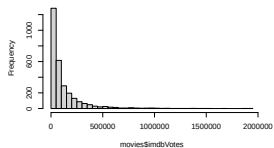
$$h = 2 \frac{\text{IQR}(x)}{\sqrt[3]{n}}$$

Doing in R

Basic R functionality for histogram allows to directly state the method for bin calculations

```
par(mfrow = c(2,2))  
hist(movies$imdbVotes, breaks = 'sturges', main = 'Sturges', )  
hist(movies$imdbVotes, breaks = 'fd', main = 'Freedman-Diaconis')  
hist(movies$imdbVotes, breaks = 'scott', main = 'Scott')
```

Doing in R

Sturges**Freedman-Diaconis****Scott**

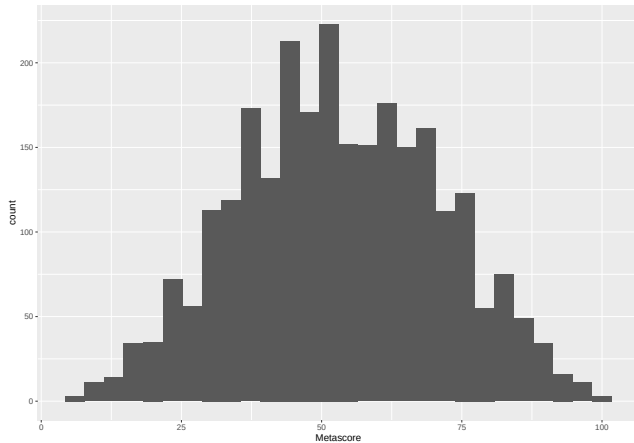
Doing in ggplot2

In ggplot2 you need to calculate the number of the bins/binwidth then provide the result as an argument

Example: Rice rule

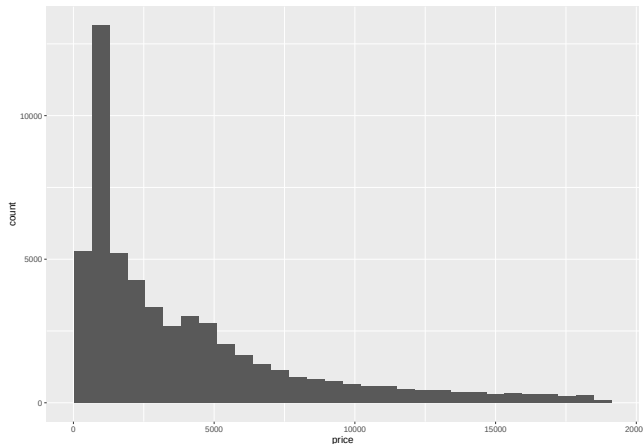
```
n <- length(movies$Metascore[!is.na(movies$Metascore)])  
k <- ceiling(2*(n^(1/3)))  
ggplot(movies, aes(x = Metascore)) + geom_histogram(bins = k)
```


Doing in ggplot2



Bumpy distribution

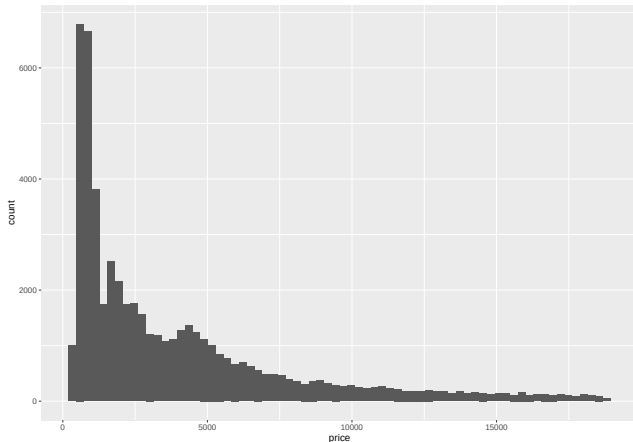
```
data("diamonds")  
ggplot(diamonds, aes(price)) + geom_histogram()
```



Bumpy distribution

Increase the number of bins, the bump becomes more apparent

```
ggplot(diamonds, aes(price)) + geom_histogram(bins = 70)
```



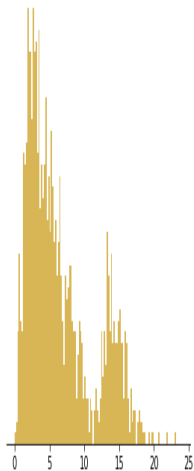
Bumpy distribution

How can you detect it ?

bin size = 0.2

bin size = 1.0

bin size = 5.0

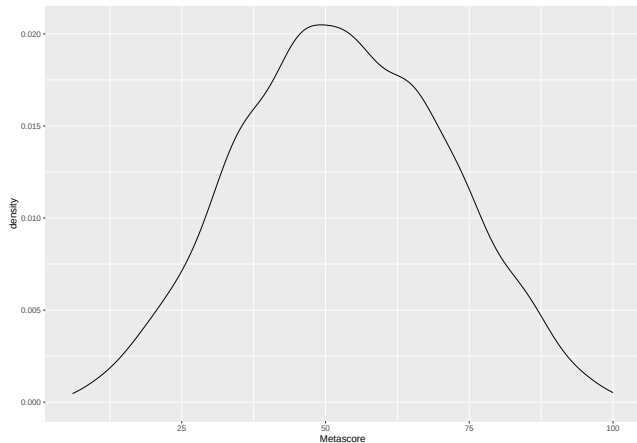


Density plots

- To visualize the distribution of the continuous variable, you can also use kernel density estimate - smoothed version of the histogram.
- In ggplot it is done with the `geom_density()`

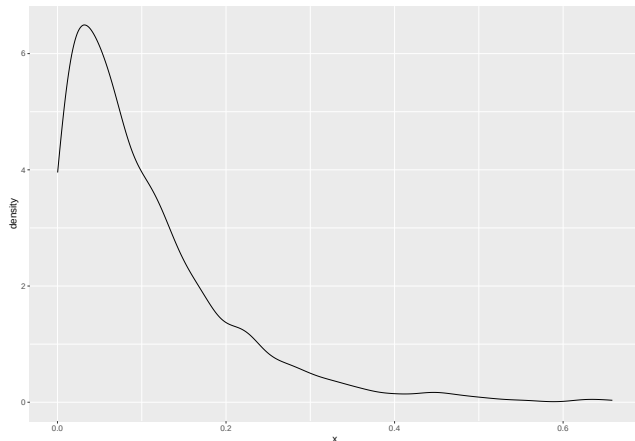
Density plots

```
ggplot(movies, aes(x = Metascore)) + geom_density()
```



Density plots: exponential distribution

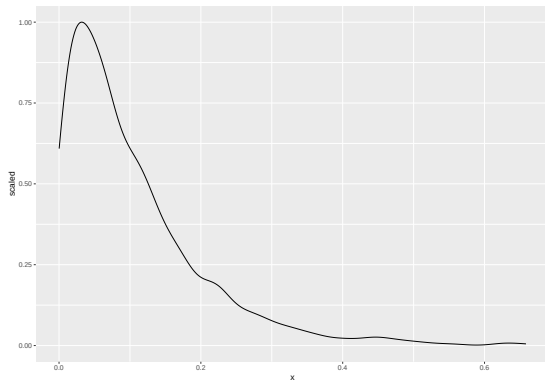
```
set.seed(1)
x <- rexp(1000, rate = 10)
ggplot() + geom_density(aes(x))
```



Density plots

Scale the y axis

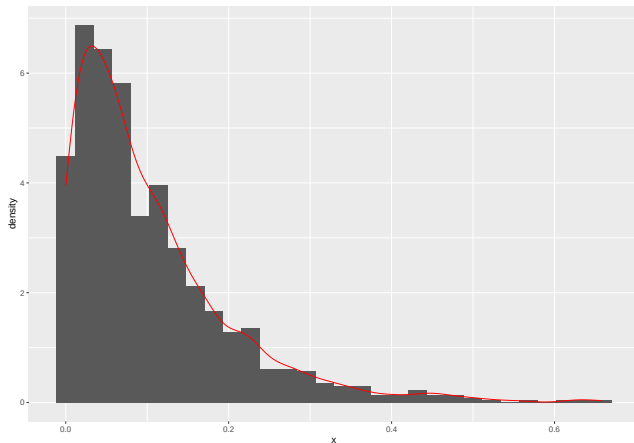
```
ggplot() + geom_density(aes(x, y = after_stat(scaled)))
```



Note: We use `after_stat(scaled)` instead of the deprecated `..scaled..` syntax (changed in ggplot2 3.4.0+).

Density plot over the histogram

```
ggplot(mapping = aes(x = x)) + geom_histogram(aes(y = after_stat(density)))  
  geom_density(color = 'red')
```



Note: `after_stat(density)` replaces the deprecated `..density..` notation. This makes it explicit that density is computed by the stat layer.

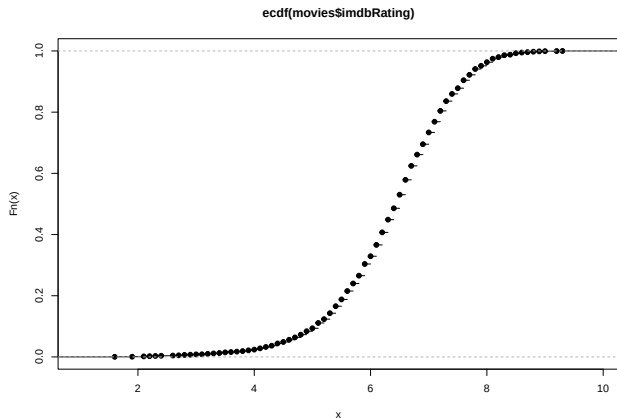
Cumulative distribution function

- The cumulative distribution function (CDF) of a random variable X , or just distribution function of X , evaluated at x is the $P(X \leq x)$.
- When we have the data, we have the empirical distribution, we can construct Empirical Cumulative Distribution Function:

$$\hat{F}_n(x) = \frac{\text{number of elements in the sample} \leq x}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{I}(X_i \leq x)$$

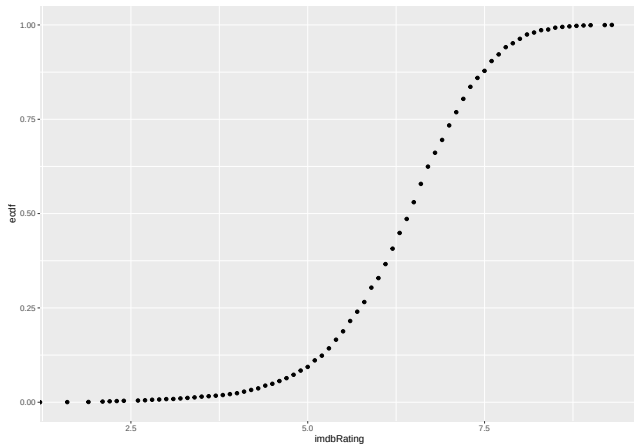
eCDF

```
plot(ecdf(movies$imdbRating))
```



eCDF

```
ggplot(movies, aes(x = imdbRating)) + stat_ecdf(geom = 'point')
```

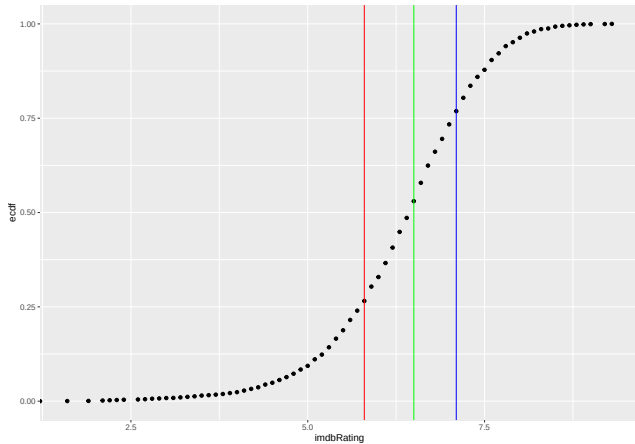


eCDF

To make eCDF more informative we can add quartiles to the plot

```
quant <- quantile(movies$imdbRating, probs = c(0.25,0.5,0.75), na.rm = T)
ggplot(movies, aes(x = imdbRating)) + stat_ecdf(geom = 'point') +
  geom_vline(xintercept = quant, color = c('red', 'green', 'blue'))
```

eCDF

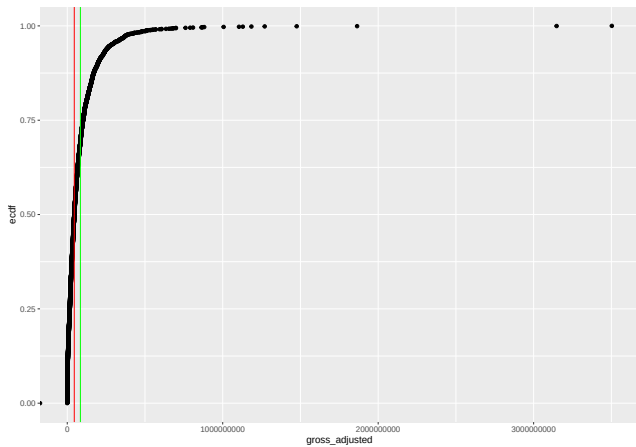


eCDF

Other distribution

```
med <- median(movies$gross_adjusted)
avg <- mean(movies$gross_adjusted)
ggplot(movies, aes(x = gross_adjusted)) + stat_ecdf(geom = 'point') +
  geom_vline(xintercept = c(med, avg), color = c('red', 'green'))
```

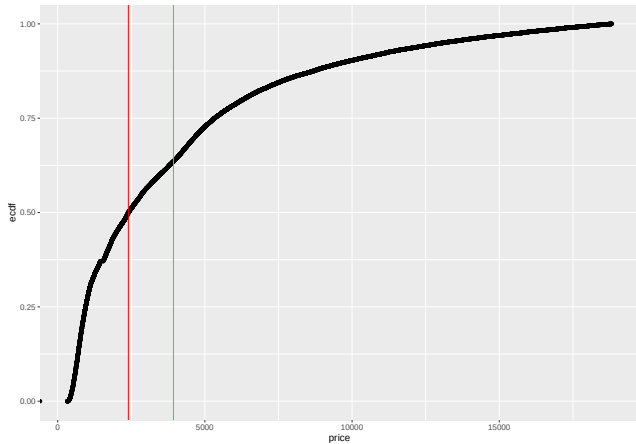
eCDF



Another skewed distribution

```
med <- median(diamonds$price)
avg <- mean(diamonds$price)
ggplot(diamonds, aes(x = price)) + stat_ecdf(geom = 'point') +
  geom_vline(xintercept = c(med, avg), col = c("red", "green"))
```

Another skewed distribution



Section 2

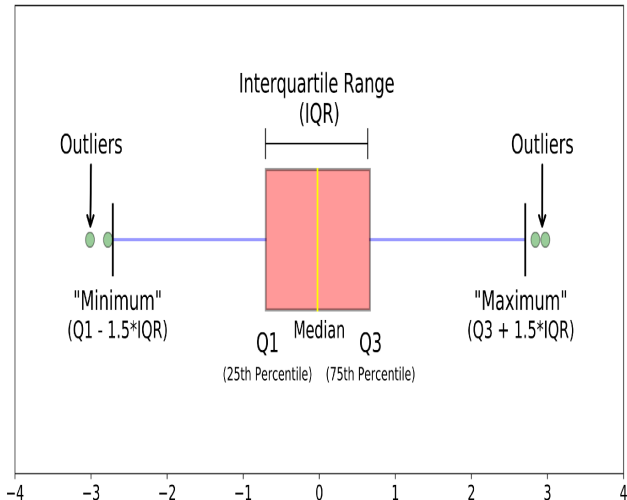
Boxplots

Boxplot

- Boxplot or box-whisker plot, is another way to display the distribution of the continuous variable
- Boxplots are usually used to visualize the distribution of some continuous variable by categories of a categorical variable
- They are also used to detect outliers (non-parametric way)

Boxplot

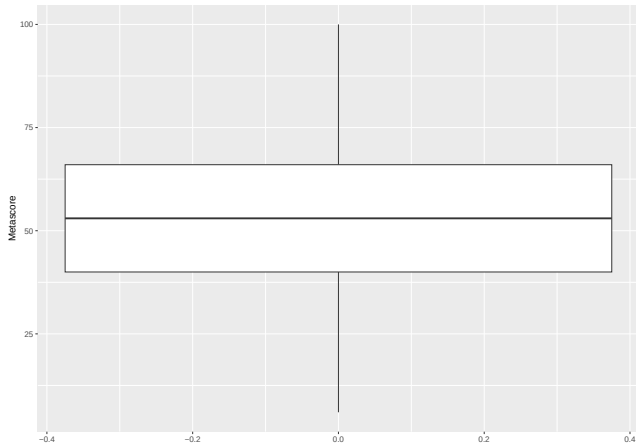
The structure of the boxplot



Boxplot

Vertical Boxplot

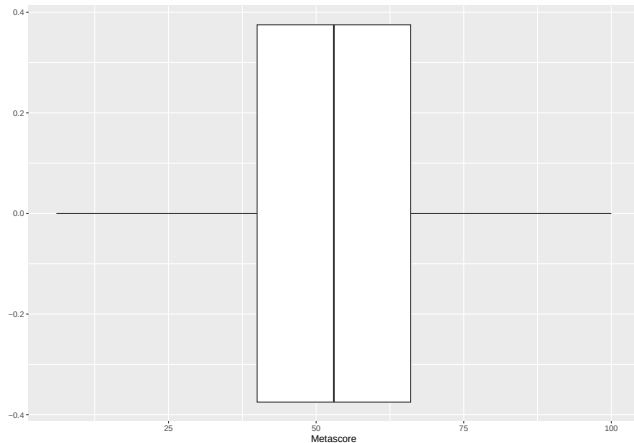
```
ggplot(data = movies, aes(y = Metascore)) + geom_boxplot()
```



Boxplot

Horizontal boxplot

```
ggplot(data = movies, aes(x = Metascore)) + geom_boxplot()
```



Boxplot

Reading the boxplot

- The width of the box - IQR, is an indicator of the variance
- If the median is in the center and the whiskers have the same length with small to none outliers, then the variable has a bell shape

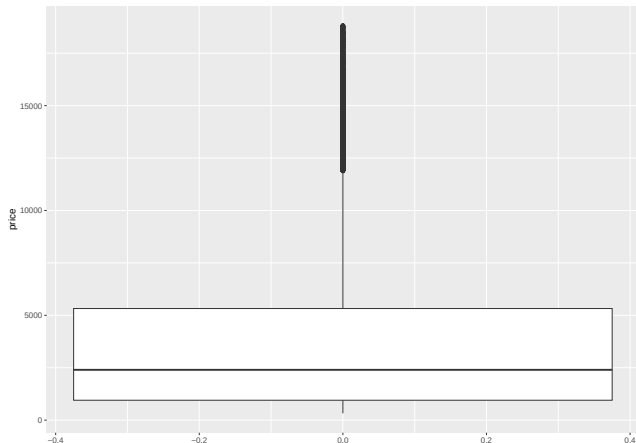
Boxplot

Boxplot of skewed distribution

- Top whisker is longer
- Outliers on the top
- Initial conclusion: We have right skewed distribution

Boxplot

```
ggplot(data = diamonds, aes(y = price)) + geom_boxplot()
```

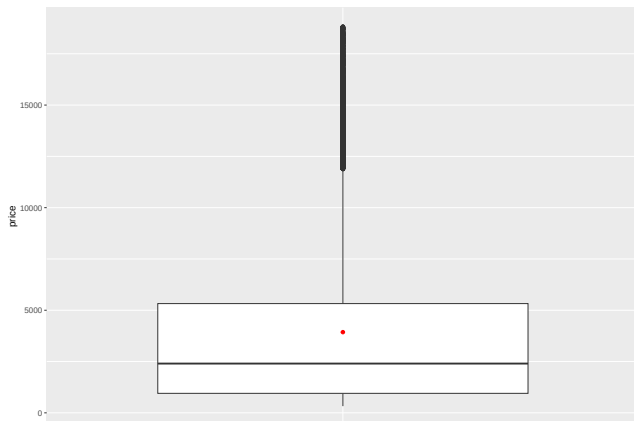


Boxplot

- You can also add mean to the graph (added empty x in aesthetics)
- Mean greater than median - right skewed distribution

Boxplot

```
ggplot(data = diamonds, aes(x = "", y = price)) + geom_boxplot() +  
  stat_summary(fun = mean, geom = 'point', color = 'red') + xlab("") +  
  theme(axis.ticks.x = element_blank())
```



Comparing distributions

Sometimes will need to compare the distribution of one continuous variable by different categories of a categorical variable

- The height of males and females
- Distribution of waiting time by weekdays
- Number of goals per position in football, etc

We can do this with histograms, boxplots and eCDF

Comparing distributions

- We will use Galton's hereditary data
- for the full analysis refer to Regression Towards Mediocrity in Hereditary Stature

Comparing Distributions

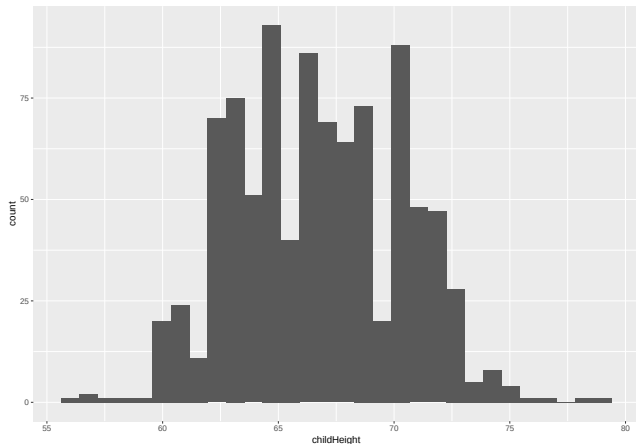
```
data("GaltonFamilies")
str(GaltonFamilies)

## 'data.frame':    934 obs. of  8 variables:
## $ family      : Factor w/ 205 levels "001","002","003",...: 1 1 1 1 2
## $ father      : num  78.5 78.5 78.5 78.5 75.5 75.5 75.5 75.5 75 75 ...
## $ mother      : num  67 67 67 67 66.5 66.5 66.5 66.5 64 64 ...
## $ midparentHeight: num  75.4 75.4 75.4 75.4 73.7 ...
## $ children     : int   4 4 4 4 4 4 4 4 2 2 ...
## $ childNum     : int   1 2 3 4 1 2 3 4 1 2 ...
## $ gender       : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 1 1
## $ childHeight  : num  73.2 69.2 69 69 73.5 72.5 65.5 65.5 71 68 ...
```

Comparing distributions

Histogram of height

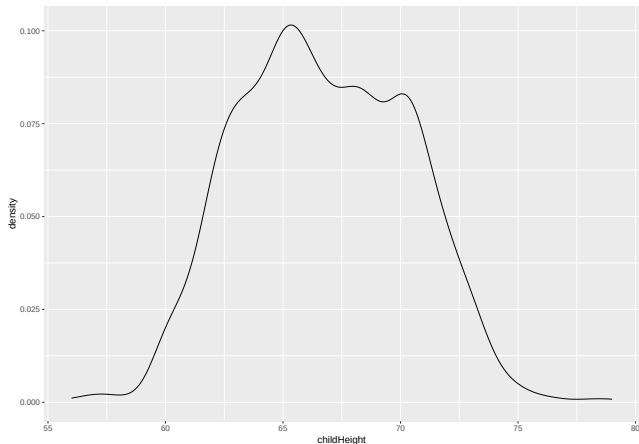
```
ggplot(data=GaltonFamilies, aes(x = childHeight)) + geom_histogram()
```



Comparing distributions

Density estimate

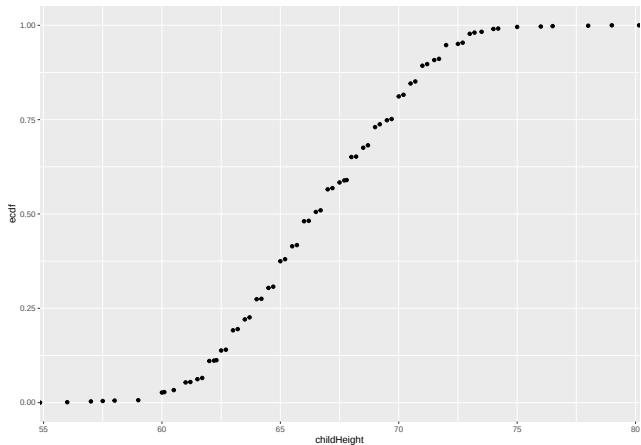
```
ggplot(data = GaltonFamilies, aes(x = childHeight)) + geom_density(bw = 0.7
```



Comparing distributions

eCDF

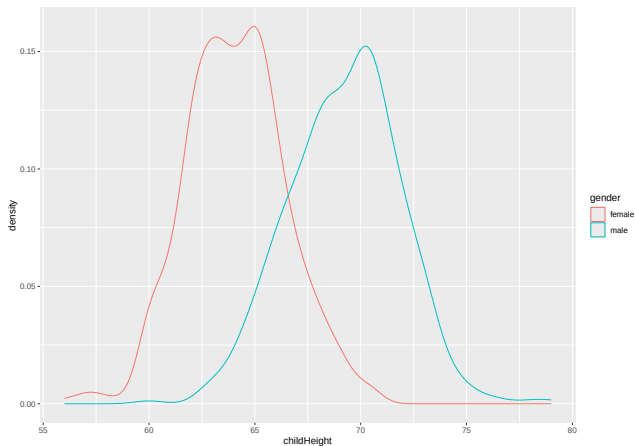
```
ggplot(data = GaltonFamilies, aes(childHeight)) + stat_ecdf(geom = 'point')
```



Comparing distributions

Compare height for male and female children There are two distinct distributions

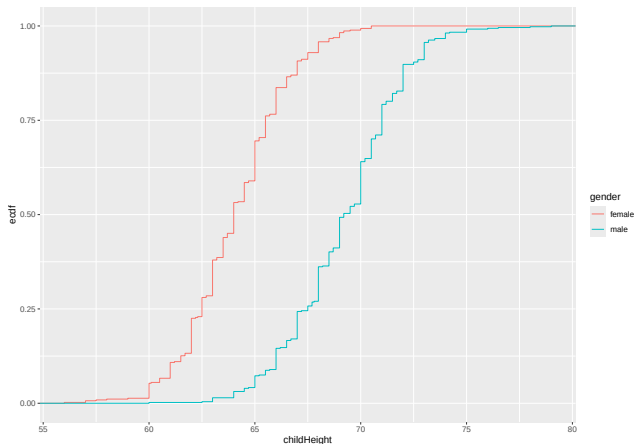
```
ggplot(data = GaltonFamilies, aes(x = childHeight, color = gender)) +  
  geom_density()
```



Comparing distributions

Create the eCDF

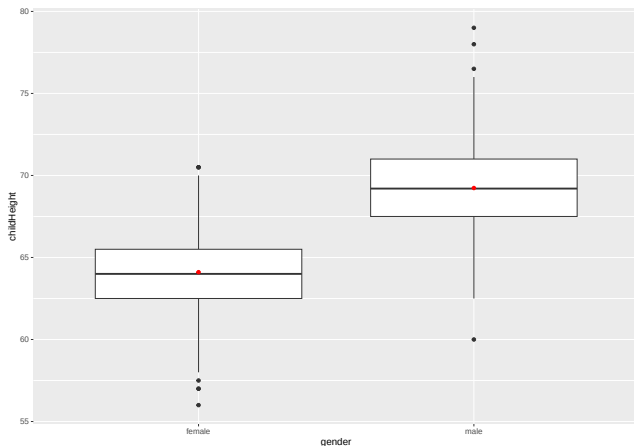
```
ggplot(data = GaltonFamilies, aes(x = childHeight, color = gender)) +  
  stat_ecdf()
```



Comparing distributions

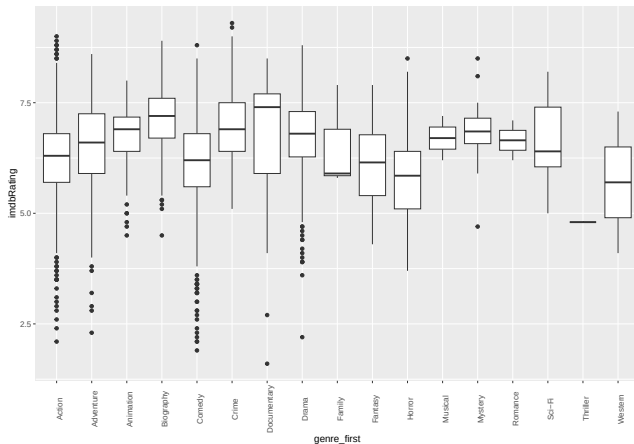
Boxplot

```
ggplot(data = GaltonFamilies, aes(x = gender, y = childHeight)) +  
  geom_boxplot() +  
  stat_summary(fun.y = mean, geom = 'point', color = 'red')
```



Comparing distributions

```
ggplot(movies, aes(x = genre_first, y = imdbRating)) + geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90))
```



Section 3

Testing for distribution

Probability plots

- The probability plot is a graphical technique for assessing whether or not a variable follows a given distribution.
- The data is plotted against a theoretical distribution in such a way that the points should form approximately a straight line.
- Departures from this straight line indicate departures from the specified distribution.

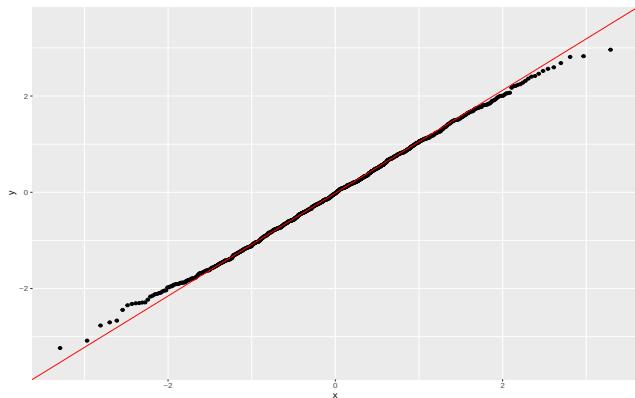
q-q plots

- Q-Q Plots (Quantile-Quantile plots) are plots of quantiles of two variables plotted against each other.
- A quantile is a fraction where certain values fall below that value.
- The purpose of q-q plots is to find out if two sets of data come from the same distribution.
- If we have standard normal distribution then 45° angle is plotted on the QQ plot; if the two data sets come from the same distribution, the points will fall on that reference line, thus quantiles of theoretical and sample distributions are the same.
- The greater the departure from the reference line, the greater the evidence for the conclusion that the two data sets have come from populations with different distributions.
- If the data does not follow standard normal distribution, then the reference line is formed with intercept = mean and slope = standard deviation

q-q plots

We have 45° line

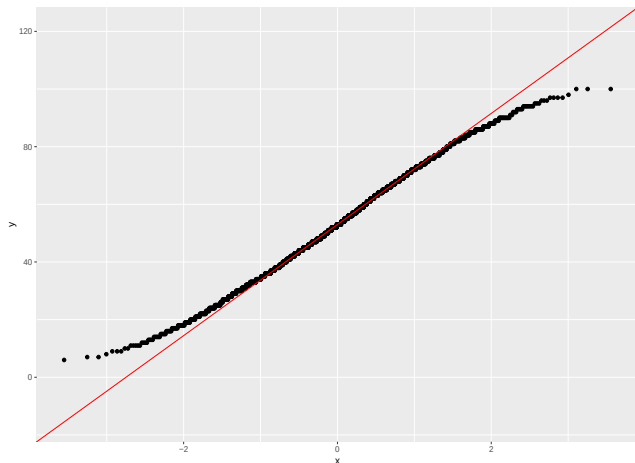
```
x <- rnorm(1000)
ggplot(mapping = aes(sample = x)) + geom_qq() +
  geom_qq_line(color = 'red')
```



q-q plots

q-q plot for Metascore

```
ggplot(movies, aes(sample = Metascore)) + geom_qq() +  
  geom_qq_line(color = 'red')
```



q-q plots

Pay Attention

- We have corresponding Z scores for theoretical values on x axis
- The quantiles from the sample in the original scale
- Line is not 45° any more, but is rather estimated

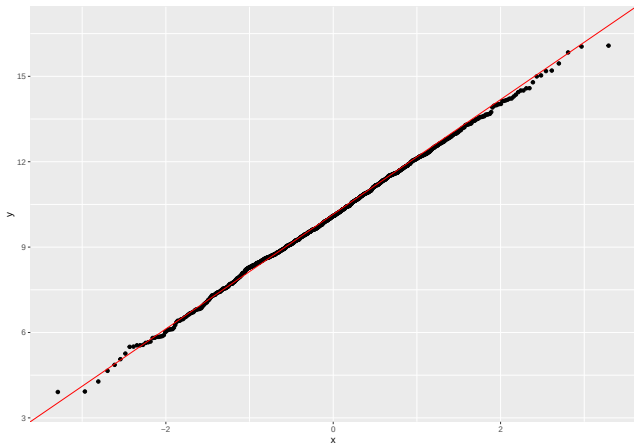
q-q plots

Lets see what happens when you define the distribution parameters

```
x <- rnorm(1000, mean=10, sd = 2)
p1 <- ggplot(mapping = aes(sample = x)) + geom_qq() +
  geom_qq_line(color = 'red')
```

q-q plots

p1



q-q plots

Get the data used to draw the plot

```
df1 <- ggplot_build(p1)$data[[2]]
df1
```

##		x	y	slope	intercept	PANEL	group	colour	linewidth	line
## 1		-3.290527	3.52676	2.01425	10.1547	1	-1	red	0.5	
## 2		3.290527	16.78265	2.01425	10.1547	1	-1	red	0.5	
##		alpha								
## 1		NA								
## 2		NA								

q-q plots

Calculate the slope

```
slope <- diff(df1$y)/diff(df1$x)
slope
## [1] 2.01425
```

Intercept

```
df1$y[1] - slope*df1$x[1]
## [1] 10.1547
```

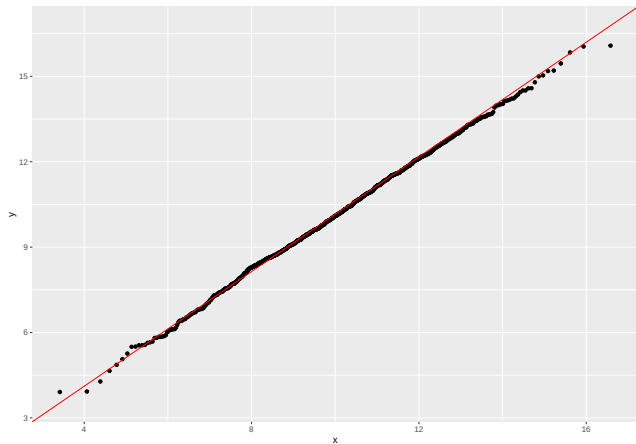

q-q plots

Define the right scale for theoretical distribution

```
p1 <- ggplot(mapping = aes(sample = x)) +  
  geom_qq(dparams = list(mean = 10, sd = 2)) +  
  geom_qq_line(color = 'red', dparams = list(mean = 10, sd = 2))
```

q-q plots

p1



q-q plots

Calculate the slope

```
df1 <- ggplot_build(p1)$data[[2]]  
  
slope <- diff(df1$y)/diff(df1$x)  
slope  
## [1] 1.007125
```

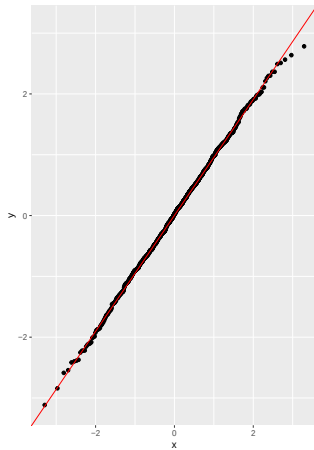
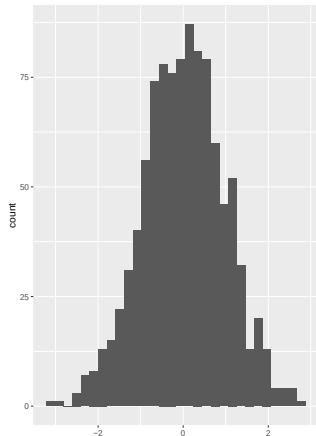
Intercept

```
df1$y[1] - slope*df1$x[1]  
## [1] 0.08345323
```

q-q plots

Normal distribution

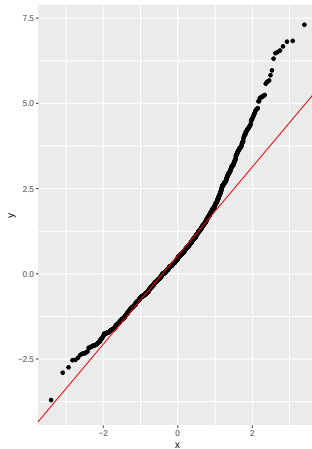
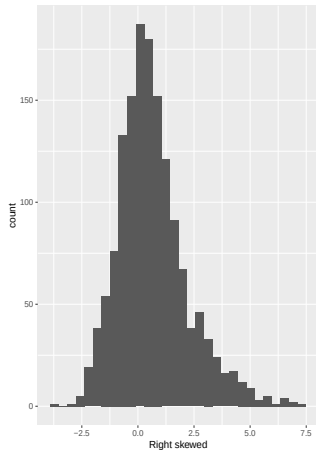
Symmetric distribution



q-q plots

Right skewed distribution

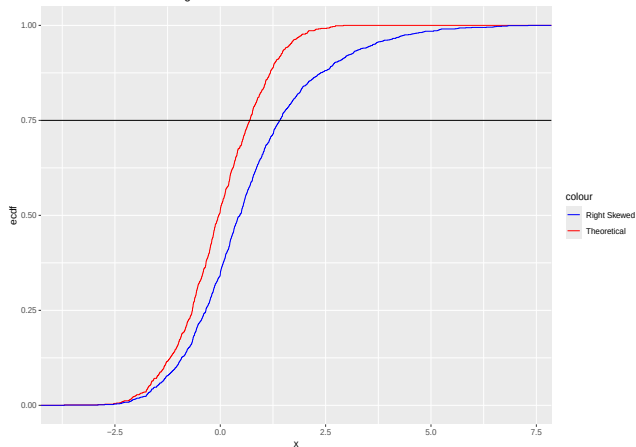
Right skewed distribution



q-q plots

eCDF for Right skewed and Theoretical distributions

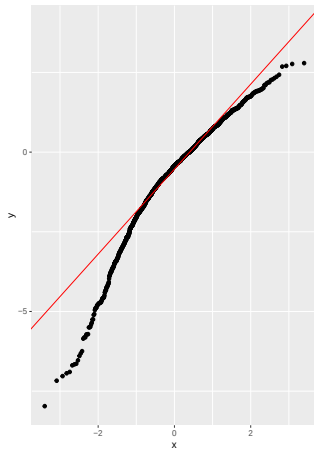
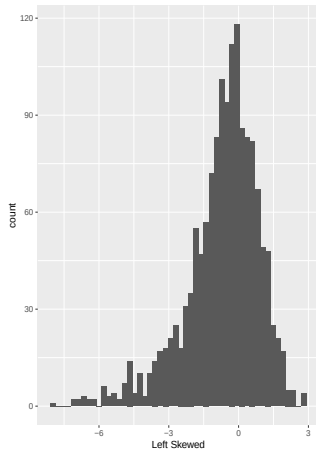
eCDF for Theoretical and Right skewed distribution



q-q plots

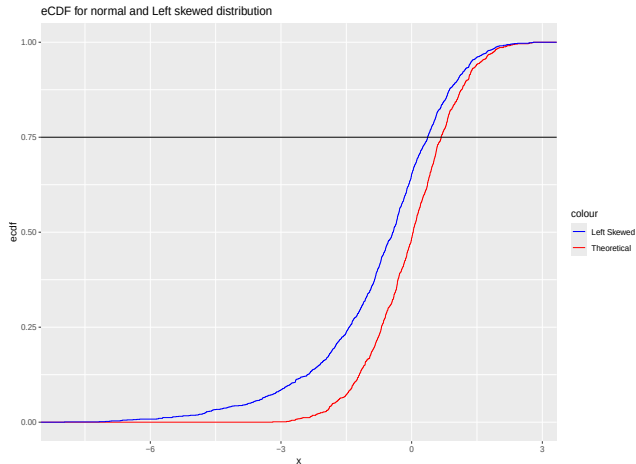
Left skewed distribution

Left skewed distribution



q-q plots

eCDF for left skewed distribution



q-q plots

- Right skewed distribution: The points' upward trend shows that the sample quantiles are much greater than the theoretical quantiles.
- Left skewed distribution: The sample quantiles are going to be much lower than the theoretical quantiles.

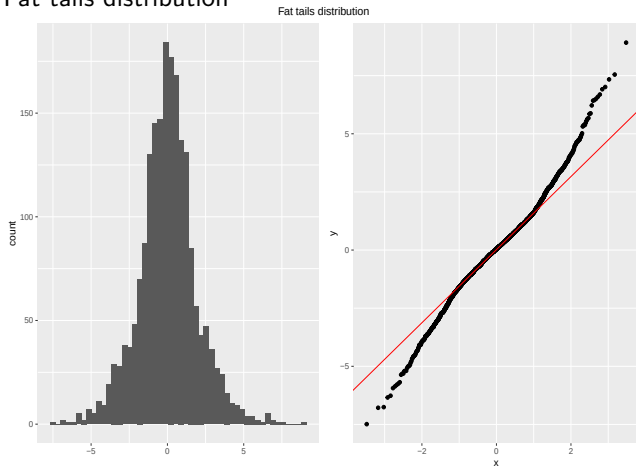
q-q plots

Heavy tail (Fat tail)

- Fat tail means that compared to the normal distribution there is more data located at the extremes of the distribution and less data in the center of the distribution.
- In terms of quantiles this means that the first quantile is much less than the first theoretical quantile and the last quantile is greater than the last theoretical quantile

q-q plots

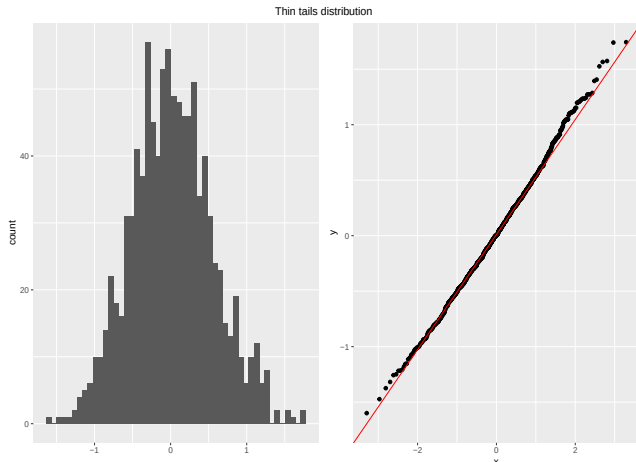
Fat tails distribution



q-q plots

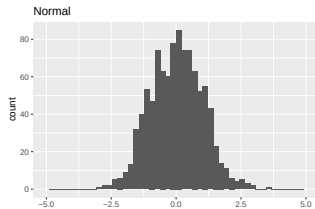
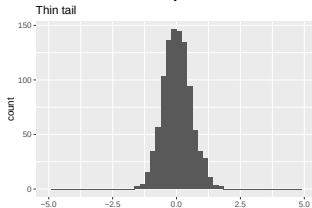
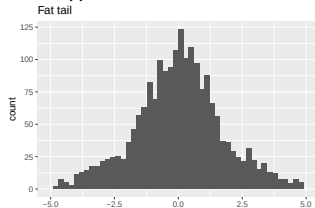
Thin tails distribution

With thin tails distribution you have less data in the tails than it should be compared to the normal distribution



q-q plots

Setting the scale for X axis the same for all plots



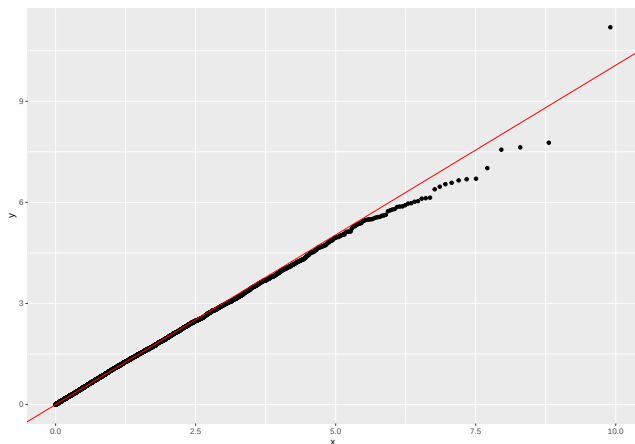
q-q plots

- By default the Q-Q plot is constructed with normal distribution as a theoretical distribution
- However you can use any distribution that is available in R
- Or provide your own distribution with parameters

q-q plots

q-q plot for exponential distribution

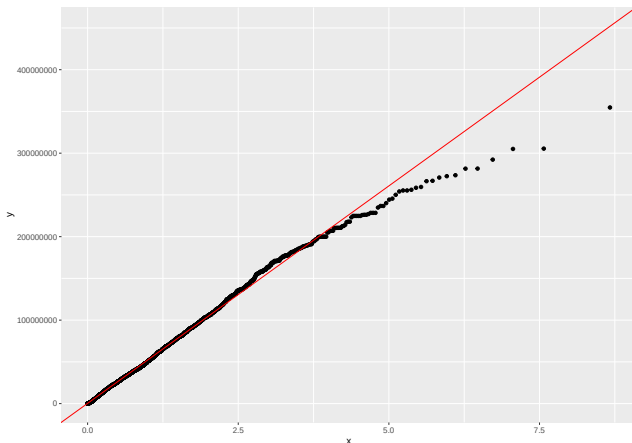
```
x_exp <- rexp(10000)
ggplot(mapping = aes(sample = x_exp)) + geom_qq(distribution = stats::qexp)
  geom_qq_line(color = 'red', distribution = stats::qexp)
```



q-q plots

Budget

```
ggplot(movies, aes(sample = budget_adjusted)) +  
  geom_qq(distribution = stats::qexp) +  
  geom_qq_line(color = 'red', distribution = stats::qexp)
```



Section 4

Aligning the bins

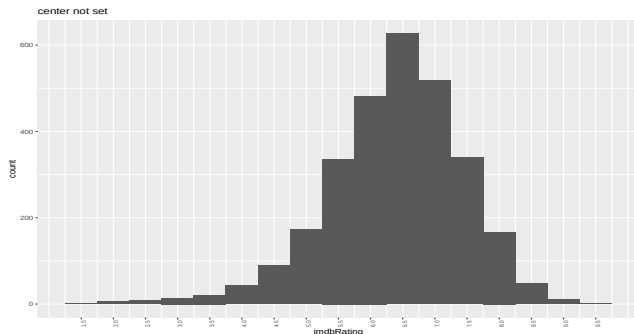
Bin center

- Bin width is not the only choice that changes a histogram
- Each bin also has a position. In `ggplot2` that position is set with `center`
- The same `binwidth` with a different center can turn one peak into a flat top, or move a bump
- Try more than one center before you describe a mode
- If `center` is omitted, `ggplot` still chooses an alignment.

Bin center

imdbRating, binwidth = 0.5, and no center. The ticks mark the bin centers.

```
p_center_default <- ggplot(movies, aes(x = imdbRating)) +
  geom_histogram(binwidth = 0.5) +
  ggtitle('center not set')
p_center_default +
  scale_x_continuous(breaks = ggplot_build(p_center_default)$data[[1]]$x) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7))
```



Bin center

No center was passed. ggplot puts a boundary at half the bin width, here 0.25.

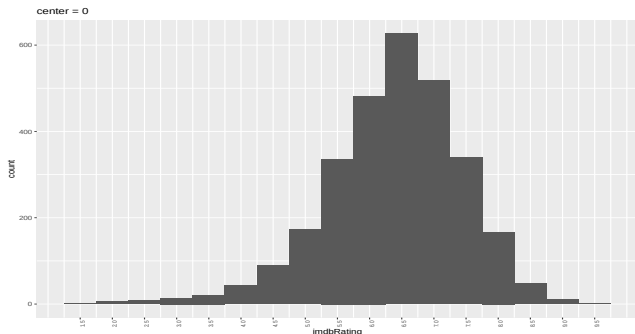
```
bins_default <- ggplot_build(p_center_default)$data[[1]]
knitr::kable(bins_default[, c('xmin', 'x', 'xmax', 'count')], linesep = '')
```

xmin	x	xmax	count
1.25	1.5	1.75	1
1.75	2.0	2.25	6
2.25	2.5	2.75	8
2.75	3.0	3.25	14
3.25	3.5	3.75	21
3.75	4.0	4.25	44
4.25	4.5	4.75	90
4.75	5.0	5.25	172
5.25	5.5	5.75	336
5.75	6.0	6.25	482
6.25	6.5	6.75	627
6.75	7.0	7.25	518
7.25	7.5	7.75	340
7.75	8.0	8.25	167
8.25	8.5	8.75	47
8.75	9.0	9.25	10
9.25	9.5	9.75	1

Bin center

imdbRating, binwidth = 0.5, center = 0

```
p_center0 <- ggplot(movies, aes(x = imdbRating)) +
  geom_histogram(binwidth = 0.5, center = 0) +
  ggtitle('center = 0')
p_center0 +
  scale_x_continuous(breaks = ggplot_build(p_center0)$data[[1]]$x) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7))
```



Bin center

`xmin` and `xmax` are the edges, `x` is the center. For this bin width these edges match the default. The tallest bar is the single bin at 6.5.

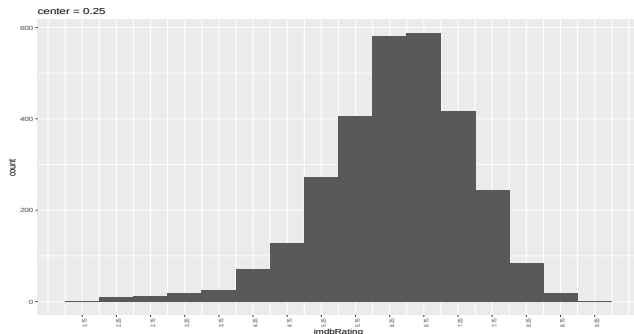
```
bins0 <- ggplot_build(p_center0)$data[[1]]
knitr::kable(bins0[, c('xmin', 'x', 'xmax', 'count')], linesep = '')
```

xmin	x	xmax	count
1.25	1.5	1.75	1
1.75	2.0	2.25	6
2.25	2.5	2.75	8
2.75	3.0	3.25	14
3.25	3.5	3.75	21
3.75	4.0	4.25	44
4.25	4.5	4.75	90
4.75	5.0	5.25	172
5.25	5.5	5.75	336
5.75	6.0	6.25	482
6.25	6.5	6.75	627
6.75	7.0	7.25	518
7.25	7.5	7.75	340
7.75	8.0	8.25	167
8.25	8.5	8.75	47
8.75	9.0	9.25	10
9.25	9.5	9.75	1

Bin center

Same bin width, center = 0.25

```
p_center025 <- ggplot(movies, aes(x = imdbRating)) +
  geom_histogram(binwidth = 0.5, center = 0.25) +
  ggtitle('center = 0.25')
p_center025 +
  scale_x_continuous(breaks = ggplot_build(p_center025)$data[[1]]$x) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7))
```



Bin center

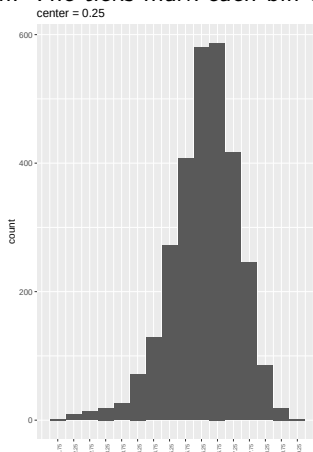
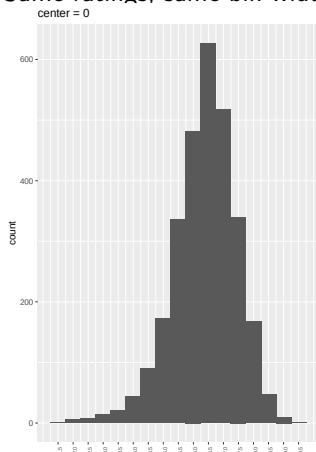
`center = 0.25` shifts every edge by 0.25. The peak is now two bins, at 6.25 and 6.75.

```
bins025 <- ggplot_build(p_center025)$data[[1]]
knitr::kable(bins025[, c('xmin', 'x', 'xmax', 'count')], linesep = '')
```

xmin	x	xmax	count
1.5	1.75	2.0	2
2.0	2.25	2.5	9
2.5	2.75	3.0	13
3.0	3.25	3.5	19
3.5	3.75	4.0	26
4.0	4.25	4.5	72
4.5	4.75	5.0	129
5.0	5.25	5.5	272
5.5	5.75	6.0	407
6.0	6.25	6.5	580
6.5	6.75	7.0	587
7.0	7.25	7.5	417
7.5	7.75	8.0	245
8.0	8.25	8.5	85
8.5	8.75	9.0	19
9.0	9.25	9.5	2

Bin center

Same ratings, same bin width. The ticks mark each bin center.



Section 5

Log scale

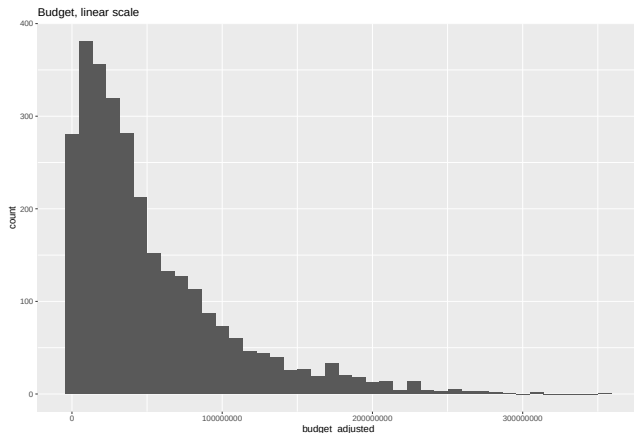
Log scale

- Budget, gross, vote counts, and diamond price pile up near zero and then stretch far to the right
- On a linear axis the histogram is a spike, and the body of the data is hard to read
- A log scale spreads that body out. Use it when every value is positive
- In `ggplot2`, add `scale_x_log10()`

Log scale

Adjusted budget on a linear axis

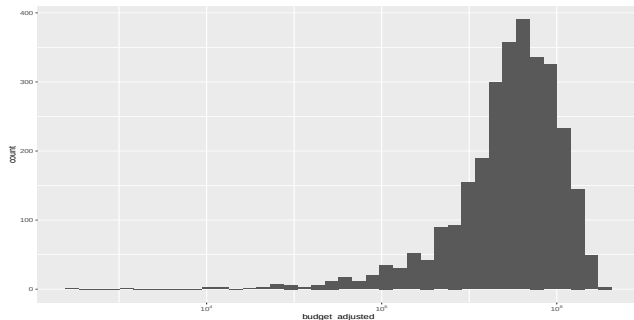
```
ggplot(movies, aes(x = budget_adjusted)) +  
  geom_histogram(bins = 40) +  
  ggtitle('Budget, linear scale')
```



Log scale

The same histogram on a log10 axis

```
ggplot(movies, aes(x = budget_adjusted)) +  
  geom_histogram(bins = 40) +  
  scale_x_log10(labels = scales::label_log()) +  
  labs(x = 'budget_adjusted')
```



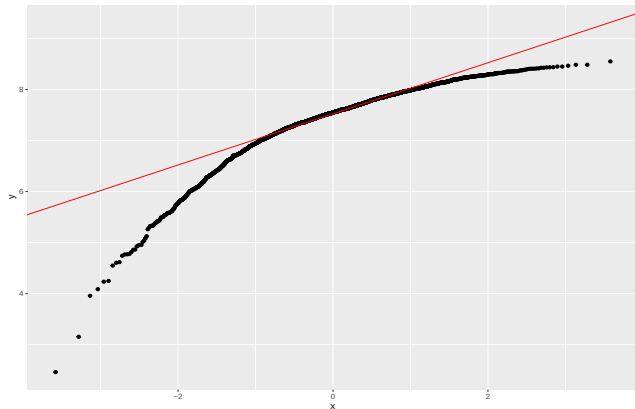
Log scale

- The log scale shows where most budgets sit. It does not make the variable normal
- If the logged values follow a normal distribution, the original variable is roughly log-normal
- Check that claim with a Q-Q plot of $\log_{10}(\text{budget_adjusted})$, not by looking at the histogram alone

Log scale

Q-Q plot of log10 budget. The ends still leave the line, so adjusted budget is not log-normal

```
ggplot(movies, aes(sample = log10(budget_adjusted))) +  
  geom_qq() +  
  geom_qq_line(color = 'red')
```



Section 6

Log-normal distribution

Log-normal distribution

- A positive variable is log-normal when its logarithm is close to a normal distribution
- On the original scale the histogram is right-skewed: most values are small, and a few are very large
- Taking $\log(x)$ or $\log_{10}(x)$ removes that skew. The two logs change only the axis labels, not the shape
- A right skew alone is not enough. Adjusted budget stayed skewed after the log. The next example does not

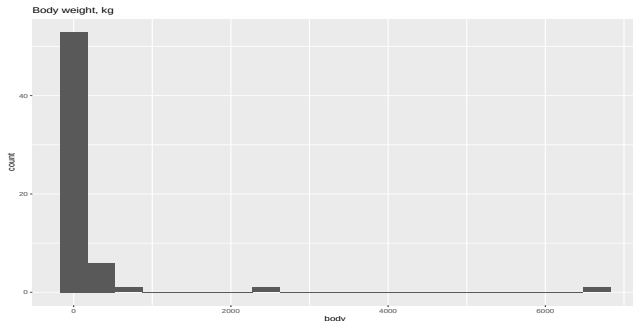
Where it is used

- Body size and other biological measurements that grow by multiplication
- Incomes, house prices, and the size of cities
- Insurance claim amounts
- The example is body weight, in kilograms, for 62 mammal species: `mammals` in the `MASS` package

Log-normal distribution

Body weight. Almost every species sits in the first bar. The two elephants stretch the axis.

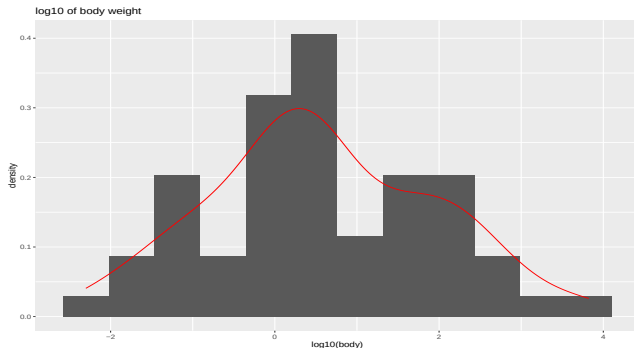
```
data(mammals)
ggplot(mammals, aes(x = body)) +
  geom_histogram(bins = 20) +
  ggtitle('Body weight, kg')
```



Log-normal distribution

$\log_{10}(\text{body})$. A value of 0 is 1 kg, and a value of 3 is 1000 kg. The curve is close to a bell.

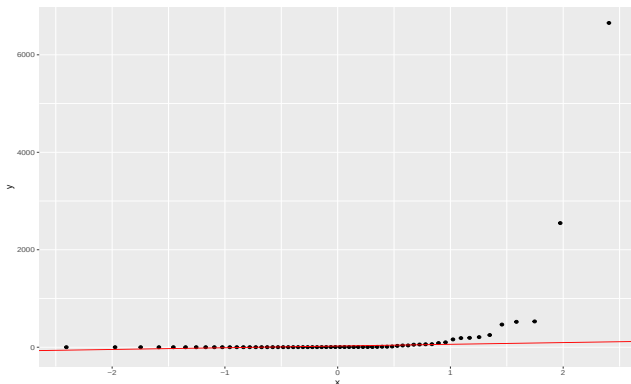
```
ggplot(mammals, aes(x = log10(body))) +  
  geom_histogram(aes(y = after_stat(density)), bins = 12) +  
  geom_density(color = 'red') +  
  ggtitle('log10 of body weight')
```



Log-normal distribution

Q-Q plot of body weight. The largest species sit far above the normal line.

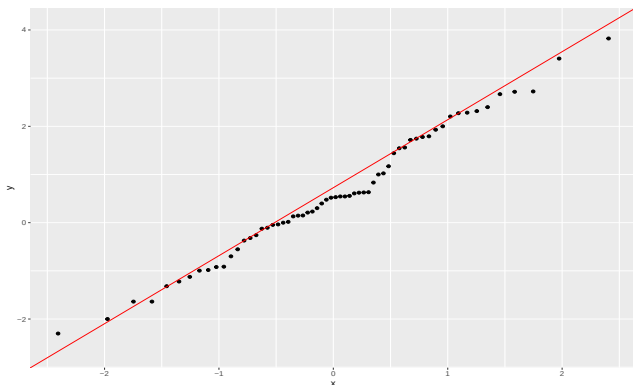
```
ggplot(mammals, aes(sample = body)) +  
  geom_qq() +  
  geom_qq_line(color = 'red')
```



Log-normal distribution

Q-Q plot of $\log_{10}(\text{body})$. The points stay near the line, so body weight is roughly log-normal.

```
ggplot(mammals, aes(sample = log10(body))) +  
  geom_qq() +  
  geom_qq_line(color = 'red')
```



A typical value

The mean of the log, raised back with $10^{\hat{}}$, is the geometric mean. It is a typical weight. The ordinary mean is the total weight shared equally, so the largest species pull it up.

```
c(mean = mean(mammals$body),
  median = median(mammals$body),
  geometric_mean = 10 ^ mean(log10(mammals$body)))
```

##	mean	median	geometric_mean
##	198.789984	3.342500	3.809656

A typical value

- The geometric mean is about 3.8 kg, and the median is about 3.3 kg. The ordinary mean is about 199 kg
- Quote the geometric mean, or the median, when the question is what a typical species weighs
- Quote the ordinary mean when the question is the total, because the heavy species really do add kilograms
- The same choice comes up for prices and incomes: a typical value from the log, and the original scale when the sum is the quantity you need

Interpreting the two means

- The ordinary mean is the arithmetic mean: add the 62 weights and divide by 62
- The geometric mean is the arithmetic mean of the logs, brought back with $10^{\cdot}$. It is also the 62nd root of the product of the weights
- Both are centers. The arithmetic mean gives each kilogram the same vote. The geometric mean gives each species the same vote on the log scale
- The geometric mean is defined only when every value is positive

Interpreting the two means

- A few heavy species pull the sum, so the arithmetic mean rises to about 199 kg
- The same species barely move the product. Doubling one weight shifts the geometric mean by as much as halving another
- On a right-skewed distribution the arithmetic mean is the larger of the two. Here it is about 50 times the geometric mean
- The geometric mean, about 3.8 kg, stays next to the median, about 3.3 kg. That is the center you see once the weights are on the log scale

Section 7

Violin plots

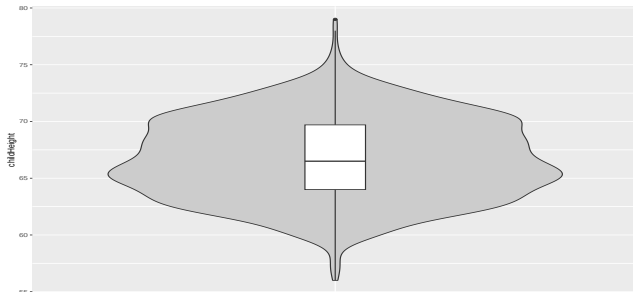
Violin plot

- A violin plot is a density estimate, drawn vertically and mirrored
- The width at a value shows how common that value is
- Use it when a boxplot is the chart you would have drawn, and the shape matters
- A boxplot reports the median and the IQR. It can hide a second peak

Violin plot

Pooled child height. The box looks symmetric. The violin has two bulges, one for each gender.

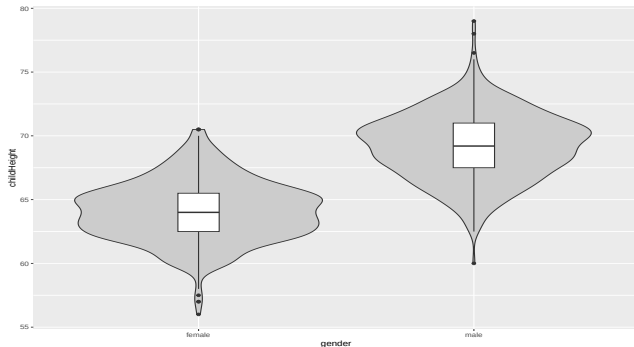
```
ggplot(GaltonFamilies, aes(x = '', y = childHeight)) +  
  geom_violin(fill = 'grey80') +  
  geom_boxplot(width = 0.12) +  
  xlab('') +  
  theme(axis.ticks.x = element_blank())
```



Violin plot

Split by gender. Each group has one peak, and the two peaks sit in different places.

```
ggplot(GaltonFamilies, aes(x = gender, y = childHeight)) +  
  geom_violin(fill = 'grey80') +  
  geom_boxplot(width = 0.15)
```



Violin plot

- Keep the boxplot inside the violin. The violin shows the shape, the box shows the median and the IQR
- Each group needs enough observations to support a smooth density
- As with a density plot, the outline can extend slightly past values that do not occur in the data

Using these tools

- If a peak appears or disappears when the bin width changes, also change center
- If a positive variable is piled on the left, such as budget, gross, votes, or price, redraw it with `scale_x_log10()`
- A log scale shows the body of the data. Confirm a log-normal claim with a Q-Q plot of the logged values
- When comparing groups, and a second peak would change the story, draw a violin with a boxplot inside